Units

⇒ Basics

⇒ Derived units with special names,

- Frequency → Hertz → Hz
- Force → Newton → N
- Pressure → Pascal → Pa
- stress → Pascal → Pa
- Energy → Joule → J
- Work → Joule → J
- Heat → Joule → J
- Power → Watt → W
- Electric charge → Coulomb → C
- Amount of eletricity → Coulomb → C
- Electric potential → Volt → V
- Potential different → Volt → V
- Eletromotive force → Volt → V
- Capacitance → Farad → F
- Electric resistance → Ohm → Ω

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> Eletric conductance → Siemens → S

> Magnetic flux → Weber → wb

> Magnetic flux density → tesla → T

> Electric inductance → henry → H

> Luminous flux → lumen → lm

> Illuminous → Lux → lx

> Radioactivity → Becquerel → Bq

> Absorbed dose → Gray → Gy

> Health effect → Sievert → Sv

> Radiation → Sievert → Sv

⟹ Derived units with compound,

Area → m × m → m^2

Volume → m × m × m → m^3

Velocity → m / s → ms^{-1}

Acceleration → ms^{-1} / s → ms^{-2}

Density → kg / m^3 → kgm^{-3}

Force → $kgms^{-2}$ → $kgms^{-2}$ (N)

Momentum → kg × ms^{-1} → $kgms^{-1}$

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Impulse $\rightarrow kgms^{-2} \times s \rightarrow kgms^{-1}$ (Ns)

Pressure $\rightarrow kgms^{-2}/m^2 \rightarrow kgm^{-1}s^{-2}$ (Pa)

Frequency $\rightarrow 1/s \rightarrow s^{-1}$ (Hz)

Work / energy $\rightarrow kgms^{-2} \times m \rightarrow kgm^2s^{-2}$ (Nm)(J)

Torque $\rightarrow kgms^{-2} \times m \rightarrow kgm^2s^{-2}$ (Nm)
(never expressed in "J")

Power $\rightarrow kgm^2s^{-2}/s \rightarrow kgm^2s^{-3}$

Moment of inertia $\rightarrow kg \times m^2 \rightarrow kgm^2$

Angular velocity $\rightarrow rad/s \rightarrow rads^{-1}$

Angular acceleration $\rightarrow rads^{-1}/s \rightarrow rads^{-2}$

Angular momentum $\rightarrow kgms^{-1} \times m \rightarrow kgm^2s^{-1}$

Universal gravitational constant $\rightarrow kgms^{-2} \times m^2/kg^2$
$\rightarrow kg^{-1}m^3s^{-2}$

Coefficient of viscosity $\rightarrow kgms^{-2} \times m/ms^{-1} \times m^2$
$\rightarrow kgm^{-1}s^{-1}$

Young's Modules $\rightarrow kgms^{-2} \times m/m^2 \times m \rightarrow kgm^{-1}s^{-2}$

Coefficient of surface tension $\rightarrow kgms^{-2}/m$
$\rightarrow kgs^{-2}$

stress $\rightarrow kgms^{-2}/m^2 \rightarrow kgm^{-1}s^{-2}$

Plank's Constant $\rightarrow kgm^2s^{-2}/s^{-1} \rightarrow kgm^2s^{-1}$

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Velocity gradient → ms^{-1}/m → s^{-1}

Pressure gradient → $kgm^{-1}s^{-2}/m$ → $kgm^{-2}s^{-2}$

⇒ Physical quantities that share the same SI units.

1. Momentum ⟷ Impulse
2. Young's modulus ⟷ Pressure
3. Plank's constant ⟷ Angular momentum
4. Torque ⟷ Work done
5. Stress ⟷ Pressure
6. Frequency ⟷ Velocity gradient
7. Angular momentum ⟷ Plank's constant

⇒ Other units Allowed is SI System.

i) Time → minute → min
hour → h
day → d

ii) Plane angle → degree → ...°
minutes → ...'
seconds → ...''

iii) Volume -> liter -> l

iv) Mass -> Ton -> t

v) Energy -> electron volt -> eV

vi) Atomic mass -> Atomic mass unit -> u

vii) length -> astronomical unit -> Pc
(Parsec)

=> Prefixes

-> Multiples

1. E (Exa) -> 10^{18}
2. P (Peta) -> 10^{15}
3. T (Tera) -> 10^{12}
4. G (Giga) -> 10^{9}
5. M (Mega) -> 10^{6}
6. k (kilo) -> 10^{3}
7. h (hecto) -> 10^{2}
8. da (deca) -> 10^{1}

-> Sub-Multiples

1. a (atto) -> 10^{-18}
2. f (femto) -> 10^{-15}
3. p (pico) -> 10^{-12}
4. n (nano) -> 10^{-9}
5. μ (micro) -> 10^{-6}
6. m (milli) -> 10^{-3}

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7. c (centi) -> 10^{-2}

8. d (deci) -> 10^{-1}

-> Relationship between cm^2 and m^2

$$1\,cm^2 = 1 \times 10^{-4}\,m^2$$

-> Relationship between cm^3 and m^3

$$1\,cm^3 = 1 \times 10^{-6}\,m^3$$

-> Relationship between mm^2 and m^2

$$1\,mm^2 = 1 \times 10^{-6}\,m^2$$

-> Relationship between mm^3 and m^3

$$1\,mm^3 = 1 \times 10^{-9}\,m^3$$

=> Dimensions are symbolized as follows.

1) [Mass] -> M

2) [Length] -> L

3) [Time] -> T

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→ Physical quantities without dimensions.

i) Plane angle
ii) Solid angle

But... have units.

i) Relative index
ii) Relative density
iii) Relative humidity
iv) Relative permitivity
v) Relative permiability
vi) Coefficient of friction
vii) Strain
viii) Efficiency
ix) Mechanical advantage
x) Velocity ratio

Also... dosen't have units.

•) Relative velocity - have units & dimensions.

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